Mobile Application and Development

Unit-I

2Marks

1.Define Mobile Computing.

Mobile Computing is a technology that provides an environment that enables users to transmit data from one device to another device without the use of any physical link or cables.

2.What are the different types of mobile Applications?

- Native apps
- Web apps
- Hybrid apps

3.Define Native App.

A native application is a software program developers build for use on a particular platform or device. Native app builders use the following programming languages to create their apps: JAVA,Python,C++, Objective-C, Kotlin, Swift, React.

Ex. Twitter, or games, such as Pokémon GO.

4.Define Web App.

A web application is a type of app that can be accessed through a web browser. When accessed in a browser on a mobile device, web apps look and behave like mobile apps — but they are not the same.

5.Define Hybrid App.

A hybrid application is a software app that combines elements of both native and web applications. They can be accessed via a web browser and downloaded from app stores.

Ex. Gmail , Instagram

6.Define Android.

Android is an operating system and programming platform developed by Google for mobile phones and other mobile devices, such as tablets.

7.What is meant by AVD?

An Android Virtual Device (AVD) is a configuration that defines the characteristics of an Android phone, tablet, Wear OS, Android TV, or Automotive OS device that you want to simulate in the Android Emulator.

8.What is the use of Breakpoints?

Once a breakpoint is reached, it will pause execution, allowing you to use other debugging tools at runtime to get an up-close look at what is happening, and what's gone wrong.

9.What is meant by conditional breakpoints?

A conditional breakpoint is a breakpoint that works only when it is relevant, that is when certain conditions are met. Conditional breakpoints are especially useful when you are trying to find bugs that are only considered bugs under certain conditions.

5 Marks

1.Explain about Native Applications.
2.Explain about Hybrid Applications.
3.Explain types of Mobile applications.
4.Discuss about mobile front-end framework.
5.Explain about AVD.
6.How to set Breakpoints?
7.How to publish Android Application.

10 Marks

1.Compare Native vs Hybrid Applications.
2.Explain Mobile Application Development Life cycle.
3.Explain Frontend vs backend for mobile app development.
4.Expalin in detail about Android Architecture.
5.How to launch your First Android Application?
6.Explain in detail about debugging android applications.

Unit-II

2Marks

1.List the building blocks of an Android application.

Activities, views, intents, services, content providers, fragments and AndroidManifest.xml.

2.Define Activity.

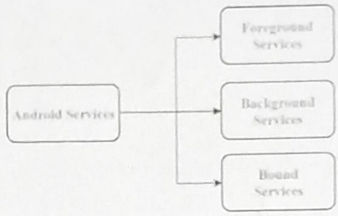
An activity represents a single screen with a user interface just like window or frame of Java. Android activity is the subclass of ContextThemeWrapper class.

3.What is the difference between intent and intent filter?

Intents are used to start new activity from the current activity. Intent filter is used for broadcast receiver. Whenever the intent filter condition is matching the android OS will launch that activity.

4.Define Service.

Services in Android are a special component that facilitates an application to run in the background in order to perform long-running operation tasks.



5.What is the purpose of the content provider component ?

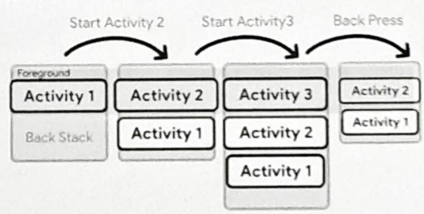
A content provider component supplies data from one application to others on request. Such requests are handled by the methods of the ContentResolver class. A content provider can use different ways to store its data and the data can be stored in a database, in files, or even over a network.

6.Brief about AndroidManifest.xml file.

Every project in Android includes a Manifest XML file, which is AndroidManifest.xml, located in the root directory of its project hierarchy. The manifest file is an important part of our app because it defines the structure and metadata of our application, its components, and its requirements.

7.Elaborate the role of the 'back stack'.

When doing a job, users engage with a task, which is a set of actions. The activities are stacked in the order in which they are opened in a stack called the back stack.



8.Define Fragments.

A Fragment is a piece of an activity which enable more modular activity design. It will not be wrong if we say, a fragment is a kind of sub-activity.

9.What is the main difference between activity and a fragment.

Activity	Fragment
Activity is an application component that gives a user interface where the user can interact.	The fragment is only part of an activity, it basically contributes its UI to that activity.
Activity is not dependent on fragment	Fragment is dependent on activity. It cannot exist independently.
we need to mention all activity it in the manifest.xml file	Fragment is not required to mention in the manifest file
Activity can exist without a Fragment	Fragment cannot be used without an Activity.

10.Define Notification.

A notification is a message that Android displays outside your app's UI to provide the user with reminders, communication from other people, or other timely information from your app.

5 Marks

1.Describe the elements of AndroidManifest.xml file.
2.How to create an activity in Android?
3.Mention the types of intents.
4.Describe the parts of a notification.
5.Explain briefly the fragment life cycle

10 Marks

1.Explain in detail about activity life cycle.

2.Explain about the onStart(), onResume(), onPause(), onStop() and onDestroy() callback functions.

3.How to create a fragment?

Unit-III

2Marks

1.Define view and viewgroup.

View is the base class for widgets, which are used to create interactive UI components like buttons, text fields, etc. The ViewGroup is a subclass of View and provides invisible container that hold other Views or other ViewGroups and define their layout properties.

2.What is the pupose of onSaveInstanceState() method?

onSaveInstanceState() is a method used to store data before pausing the activity.

3.Enumerate the techniques to handle changes in screen orientation.

- Anchoring – This is the easiest way to anchor views to the four edges of the screen. When the screen orientation changes, the views can anchor neatly to the edges.
- Resizing and repositioning – These are simple techniques to ensure that views can be handle changes in screen orientation.

4.What are the units used for measuring screen densities?

dp - Density-independent pixel. 160dp is equivalent to one inch of physical screen size.
sp - Scale-independent pixel. This is like dp and is recommended for specifying font sizes.
pt - Point. A point is defined to be 1/72 of an inch, based on the physical screen size.
px - Pixel. Corresponds to actual pixels on the screen.

5.Define Action Bar.

The app bar, also known as the action bar, is one of the most important design elements in your app's activities, because it provides a visual structure and interactive elements that are familiar to users.

6.What is the advantage of creating UI programmatically using code?

- Easier to resolve merge conflicts
- Easy to see the properties of UI elements
- Supports adding animations

7.How to set layout buttons?

- android:layout_width — Used to declare the width of View and ViewGroup elements in the layout.
- android:layout_height — Used to declare the height of View and ViewGroup elements in the layout.
- android:layout_marginLeft — Used to declare the extra space used on the left side of View and ViewGroup elements.
- android:layout_marginRight — Used to declare the extra space used on the right side of View and ViewGroup elements.
- android:layout_marginTop — Used to declare the extra space used in the top side of View and ViewGroup elements.
- android:layout_marginBottom — Used to declare the extra space used in the bottom side of View and ViewGroup elements.
- android:layout_gravity — Used to define how child Views are positioned in the layout.

8.How is ActionBar displayed when using Material theme?

When using the material themes the navigation button takes over the space previously occupied by the application icon.

9.Brief about the activity level methods for listening to UI notifications.

- **onKeyUp():** This is called when a key was released. This is not handled by any of the views inside the activity
- **onKeyDown():** This is called when a key is pressed. This is not handled by any of the views inside the activity.
- **onMenuItemSelected():** This is called when any item of the menu panel is pressed by the user.
- **onMenuOpened():** This method is called when the user opens the panel's menu.

10.Describe View Class.

Viewclass which is the superclass for all the GUI components like TextView ,EditText,Button,ImageView, ImageButton,CheckBox,RadioButton,ListView,GridView,DatePicker etc.

11.Decribe View Group class.

A ViewGroup (which is itself a special type of view) provides the layout in which you can order the appearance and sequence of views. Examples of ViewGroups include LinearLayout and FrameLayout.

12.Describe match_parent and content_wrap in view class.

match_parent is used for referring to the parent size either through layout_width or layout_height attributes. wrap_content is used to wrap the view according to its actual size. It can be used with layout_height as well as layout_width.

13.How can the ViewGroup be classified?

- Basic views — Commonly used views such as the TextView, EditText, and Button views
- Picker views — Views that enable users to select from a list, such as the TimePicker and DatePicker views
- List views — Views that display a long list of items, such as the ListView and the SpinnerView views
- Specialized fragments — Special fragments that perform specific functions

14.Describe setHour(Integer hour) method of TimePicker view.

From api level 23 we must use setHour(Integer hour). In this method there is only one parameter of integer type which is used to set the value for hours.

Ex. TimePicker simpleTimePicker=(TimePicker)findViewById(R.id.simpleTimePicker);
simpleTimePicker.setHour(5);

15.Mention the use of adapters in ListView.

An adapter bridges between UI components and the data source that fill data into UI Component. Adapter holds the data and send the data to adapter view, the view can takes the data from adapter view and shows the data on different views like as spinner, list view, grid view etc.

Commonly used adapter :

(i) Array Adapter

(ii) Base Adapter

16.What is the use of Spinner view?

Spinners provide a quick way to select one value from a set. Touching the spinner displays a dropdown menu with all other available values, from which the user can select a new one.

17.How to create special fragments?

The Fragment base class can be extended from some other subclasses of the Fragment base class to create more specialized fragments. The three subclasses of Fragment are :

- ListFragment
- DialogFragment
- PreferenceFragment

18.What are the three views used to display pictures?

- ImageView-Display an image file in application.
- ImageSwitcher-Helps to add transitions on the images.
- GridView-Displays items in two-dimensional scrolling grid.

19.Illustrate scaleType attribute of ImageView.

scaleType options are used for scaling the bounds of an image to the bounds of this view.

android:scaleType="center"

20.Describe the setAnimation() method of ImageSwitcher View.

This method is used to set the animation of the appearance of the object on the screen.

Ex : ImageSwitcher simpleImageSwitcher=(ImageSwitcher)findViewById(R.id. simpleImageSwitcher);
Animation in = AnimationUtils.loadAnimation(this,android.R.anim.slide_in_left);
simpleImageSwitcher.setInAnimation(in);

21.What are the types of menus in Android?

- Android Options Menu- The options menu is usually present in the action bar.
- Android Popup Menu-Popup Menu is a menu that appears over your view and covers your view.
- Android Contextual Menu-Contextual Menu appears when you long-press over a view element.

22.How to save user preference for saving data to files?

There are two different ways to save data in Android through Shared Preferences – One is using Activity based preferences and other is creating custom preferences.

23.Brief about canGoBack() method in Webview.

canGoBack()-This method specifies the WebView has a back history item.

24.Write short notes on SQLite.

SQLite is an open-source relational database i.e., used to perform database operations on android devices such as storing, manipulating, or retrieving persistent data from the database.

- minDistance
- listener

5 Marks

1.Explain the various parts of URI.
2.Explain the list of methods in Content Provider class.
3.What are the steps involved in displaying maps.
4.Name the two ways in which you can send SMS messages in your Android Application.
5.Explain Geocoders.

10 Marks

1.Explain in detail about Content Provider.
2. What are Android Location-Based Services?

Unit-V

2Marks

1.Which method is used to open URL connection?
URL.openConnection() is used to obtain a new HttpURLConnection and casting the result to HttpURLConnection.

2.How to post content on a web server?
To upload data to a web server, configure the connection for output using setDoOutput(true).

3.How to maintain a long-lived session?
To establish and maintain a potentially long-lived session between client and server,HttpURLConnection includes an extensible cookie manager.

4.What is the use of the DownloadImage() method?
The DownloadImage() method takes the URL of the imageto download and then opens the connection to the server using the openHttpConnection() method.It returns a Bitmap object.

5.What is the use of the DownloadText() method?
The DownloadText() method accesses the text file's URL, downloads the text file and then returns the desired string of text by opening the Http connection to the server and then uses an InputStreamReader object to read each character from the stream and save it in a String object.

6.What is meant by JSON?
JSON (Java Script Object Notation) is a straightforward data exchange format to interchange the server's data. It is a better alternative for XMI, as JASON is a light weight and structured language.

7.What is meant by IntentService?
The IntentService executes task on a single background thread ie all tasks are executed sequentially.

8.What are HTTP methods?
HttpURLConnection uses the GET method by default. Other HTTP methods OPTIONS ,HEAD , PUT , DELETE and TRACE.

9.What is the use of timer class?
To execute a block of code to be executed at a regular time interval , you can use the Timer class within your Service.

5 Marks

1.How to consume Web Services Using HTTP?
2.How to consume JSON services?
3.Explain how to bind activities to services.

10 Marks

1.How to create your own services?
2.Explain threading.

25.How do you create a database in SQLite/Brief about SQLiteOpenHelper class?

SQLiteOpenHelper(Context context, String name, SQLiteDatabase.CursorFactory factory, int version): This constructor creates an object for creating, opening, and managing the database.

26.Give the syntax to create and insert data into a table in SQLite.

mydatabase.execSQL("CREATE TABLE IF NOT EXISTS CPVALUES(Username VARCHAR,Password VARCHAR);");

mydatabase.execSQL("INSERT INTO CPVALUES('admin','admin');");

27.Brief about Cursor class with example.

Cursor class contain powerful API that allows an application to read the columns and returned the query string.

Ex: Int totalColumn=cursor.getColumnCount();

5 Marks

1.Describe the attributes of views and viewgroups.
2.What are the components of ActionBar.
3.Illustrate detecting orientation changes with code.
4.Enumerate the methods used in TimePicker view.
5.Enumerate the attributes of ListView.
6.Write short notes on Persisting Data to Files.

10Marks

1.How to create user interference programmatically using code?
2.Explain about types of layouts with examples.
3.Enumerate the common attributes of Android views.
4.Explain in detail about ListView.
5.How to create menus in Android with its types.

Unit-IV

2Marks

1.What do you mean by Shared Preferences?

SharedPreferences allows you to save and require data in the form of keys and values and provides a simple method to read and write them.

2.Define Content Providers.

Content Provider will act as a central repository to store the data of the application in one place and make that data available for different applications to access whenever it is required.

3.List the Shared Prefernces file mode mapping.

- Context.MODE_PRIVATE – default value (Not accessible outside of your application)
- Context.MODE_WORLD_READABLE – readable to other apps
- Context.MODE_WORLD_WRITEABLE – read/write to other apps

4.List the useful Content Providers.

- Browser-Stores data such as browser bookmarks,browser history and so on.
- Call Log-Stores data such as missed calls, call details and so on.
- Contact-Store contact details
- MediaStore-Store media files such as audio,video and images.
- Settings-Stores the devices settings and prefernces.

5.Define Projections.

A projection is used to translate between on screen location and geographic coordinates on the surface of the Earth.

6.Define SMS Messaging.

SMS stands for short message service. It's a protocol used for sending short messages over wireless networks.

7.List the five arguments in the send TextMessage() method.

- (String) The destination address such as the phone number.
- (String) The service center address. The value of null specifies the default service center.
- (String) The text message sent to the destination address.
- (PendingIntent) The broadcast pending intent for confirming the message is successfully sent.
- (PendingIntent) the broadcast pending intent for confirming the message is successfully delivered.

8.List the four methods of requestLocationUpdates() method.

- provider
- minTime

Unit-V

CONSUMING WEB SERVICES USING HTTP

A service is said to be consumed if it successfully fulfils the web client's request. Here the primary actor is a web client which is an application that resides on the end user's computer and connects to a server over a network. The supporting actor is a Web service which provides the remote application that the web client consumes. The Web service must be connected to the network in order to be consumed. An end user performs a task on a web client that requires consumption of a service. When an end user visits a website, uses a web application, or performs a transaction on the web, that site or application might access a Web service. The end user is not necessarily aware that a Web service is involved in the task at hand. The web client makes a request on a Web service endpoint. For example, an end user uses a web browser to visit a weather reporting website. When the website's home page loads, one or more Web services to retrieve weather-related data.

To consume a service, we use a URI connection with support for HTTP-specific features. This class allows a pattern för usage as below

I. Obtain a new HttpURLConnection by calling URLopenConnection()and casting the result to HttpURLConnection.

2. Prepare the request. The primary property of a request is its URL.Request headers may also include metadata such as credentials,preferred content types, and session cookies.

3. Optionally upload a request body. Instances must be configured with setDoOutput(true) if they include a request body. Transmit data by writing to the stream returned by URLConnection.get()OutputStream().

4. Read the response. Response headers typically include metadata such as the response body's content type and length, modified dates and session cookies. The response body may be read from the stream returned by URLConnection.getInputStream(). If the response has no body, that method returns an empty stream.

5. Disconnect. Once the response body has been read, the HttpURLConnection should be closed by calling disconnect(). Disconnecting releases, the resources held by a connection so they may be closed or reused.

CONSUMING JSON SERVICES

JSON (Java Script Object Notation) is a straightforward data exchange format to interchange the server's data .It is a better alternative for XMI, as JASON is a light weight and structured language. Android supports all the JASON classes such as JSONStringer, JSONArray, and all other forms to parse the JSON data and fetch the required information by the program.JSON is language-independent. JSON objects will contain data in the form of a key value pair. In general, JSON nodes will start with a square bracket [or with a curly bracket {. The square and curly bracket's primary difference is that the square bracket [represents the beginning of a JSON Array node. Whereas, the curly bracket ({) represents a JSONObject. So one needs to call the appropriate method to get the data. Sometimes JSON data start with [. We then need to use the getJSONArray() method to get the data. Similarly, if it starts with { , then we need to use the getJSONobject() method. The syntax of the JSON file is as following:

person.name = "Gilbert";

It can also be modified like this:

person["name"] = "Gilbert";

CREATING YOUR OWN SERVICES

The example below shows the steps to create a simple service.

1. Using Android Studio, create a new Android project and name It Services.

2. Add a new Java Class file to the project and name it MyService, Populate the MyService.java file with the code:

3. In the AndroidManifest.xml file, add the code.

4. In the main.xml file add statements replacing Text View.

5. Modify the default content of res/layout/activity_main.xml file to include two buttons in linear layout.

6. No need to change any constants in res/values/strings.xml file. Android studio take care of string values.

7. Run the application to launch Android emulator and verify the result of the changes done in the application.

Binding Activities to Services

A bound service is the server in a client-server interface. A bound service allows components (such as activities) to bind to the service, send requests, receive responses, and even perform interprocess communication (IPC). A bound service typically lives only while it serves another application component and does not run in the background indefinitely.

Creating a Bound Service

When creating a service that provides binding, you must provide an IBinder that provides the programming interface that clients can use to interact with the service. There are three ways you can define the interface:

Extending the Binder class

If your service is private to your own application and runs in the same process as the client (which is common), you should create your interface by extending the Binder class and returning an instance of it from onBind(). The client receives the Binder and can use it to directly access public methods available in either the Binder implementation or even the Service.

Here is how to set it up:

1. In your service, create an instance of Binder that either:
   i. contains public methods that the client can call
   ii. returns the current Service instance, which has public methods the client can call

iii. or, returns an instance of another class hosted by the service with public methods the client can call

2. Return this instance of Binder from the onBind() callback method.
3. In the client, receive the Binder from the onServiceConnected() callback method and make calls to the bound service using the methods provided.

Using a Messenger

If you need your interface to work across different processes, you can create an interface for the service with a Messenger.

Here is a summary of how to use a Messenger:

i. The service implements a Handler that receives a callback for each call from a client.
ii. The Handler is used to create a Messenger object (which is a reference to the Handler).
iii. The Messenger creates an IBinder that the service returns to clients from onBind().
iv. Clients use the IBinder to instantiate the Messenger (that references the service's Handler), which the client uses to send Message objects to the service.
v. The service receives each Message in its Handler—specifically, in the handleMessage() method.

Using AIDL

AIDL (Android Interface Definition Language) performs all the work to decompose objects into primitives that the operating system can understand and marshall them across processes to perform IPC. The previous technique, using a Messenger, is based on AIDL as its underlying structure.

Understanding Threading

When an application is launched in Android, it creates the primary thread of execution, referred to as the "main" thread. Most thread is liable for dispatching events to the acceptable interface widgets also as communicating with components from the Android UI toolkit. To keep your application responsive, it's essential to avoid using the most thread to perform any operation which will find yourself keeping it blocked.

Threading in Android

In Android, you will categorize all threading components into two basic categories:

Threads that are attached to an activity/fragment: These threads are tied to the lifecycle of the activity/fragment and are terminated as soon because the activity/fragment is destroyed.

Threads that aren't attached to any activity/fragment: These threads can still run beyond the lifetime of the activity/fragment (if any) from which they were spawned.

Type #1: Threading Components that Attach to an Activity/Fragment

1. ASYNCTASK

AsyncTask is the most elementary Android component for threading. It's super easy and simple to use it's also good for some basic scenarios.

AsyncTask, however, falls short if you would like your deferred task to run beyond the lifetime of the activity/fragment. The fact that even the slightest of screen rotation can cause the activity to be destroyed is simply awful! So, We Come to:

2. LOADERS

Loaders are the answer to the awful nightmare mentioned above. Loaders are great at performing in that context and they automatically stop when the activity is destroyed, even more, the sweet fact is that they also restart themselves after the activity is recreated.

Type #2. Threading Components that Do not Attach to an Activity/Fragment

1. SERVICE

Service could be thought of as a component that's useful for performing long (or potentially long) operations with no UI. Yes, you heard that right! Services don't have any UI of theirs! Service runs within the main thread of its hosting process; the service doesn't create its own thread and doesn't run during a separate process unless you specify otherwise.

Every time you create a thread, you ought to call setThreadPriority(). The system's thread scheduler gives preference to threads with high priorities, balancing those priorities with the necessity to eventually get all the work done. Generally, threads within the foreground group get around 95% of the entire execution time from the device, while the background group gets roughly 5%.